

light on/off control," says I. Hari Prasad from Omtronics.

FLYON is a lighting device being developed by Quantum Dots, which glows while floating in air without any wired connection. "It works on the amalgamation of maglev and Witricity. FLYON involves lifting forces, electromagnetic suspension, position sensing, magnetic flux, flux orientation and frequency of operation including Witricity. Witricity power sources and capture devices are specially designed magnetic resonators that efficiently transfer power over short distances via the magnetic near-field," explains T.A. Babu, managing director, Quantum Dots Technology Pvt Ltd, Quantum Dots.

Plessey announced its range of MaGIC LED die manufactured on the company's patented GaN-on-Silicon technology. The blue die is referred to in the company's press release as a blue pump for its ability to pump phosphor to a white colour range.

Overall, there seems to be a focus on improving the communications aspect of lighting as well as enabling greater precision with which these can be controlled.

What happened to the controls

A virtual home for every light bulb—that is what the future of smart lighting promises us (and delivering, too!). By having an Internet address for every bulb to call home, we gain the ability to call it up and control it as we please from anywhere across the globe through the Internet.

One innovative way to control lights is when wearable devices get in on the control game, too. Misfit is



"Touch the Lights!" Last year, over Christmas, ROHM held Touch the Lights, a participatory-style event. At the event, every time a person ran a fingertip along an installed touch display, trees and cakes on the big screen would be decorated along with it

coming up with Misfit Home Service that will let Misfit fitness trackers tell the lights when to brighten up. One use case is where the tracker detects the level of sleep of the user and uses lights to simulate a sunrise inside the bedroom.

Sensor based lighting systems are now being made smarter to fit in with human behaviour and responses. For instance, motion sensor systems that switch lights based on people entering the room can now differentiate between a pet and a person. These also do not switch off immediately when a person leaves, instead keep the light on for a few minutes as people tend to feel creepy when lights behind them shut off as they walk through a hall.

Lutron and Insteon have designed their lighting kits designed to work flawlessly with Apple's HomeKit de-

vices. This means users can control lighting by speaking to their phone or watch.

Cisco Smart + Connected City Lighting solution combines with their City Multi Sensor Node to create a light-sensory network. These standards based systems gather a wide variety of data from the environment, including levels of humidity, CO₂ and O₂, UVA and UVB light, particulate matter, motion and seismic activity, video, sound and more.

There is one thing to be wary of, however. "The problem with some app based systems is that, whenever you are activating lights through a mobile phone, you are going through servers from different countries. So full dependency is on the server located in that country. This leads to a situation where external firms retain control over home automation systems. Even big firms have systems supported by servers in certain risky countries," explains Joy Biswas, product manager, Forbix Semicon.

How is Forbix solving the problem? They had designed a system in 2007 that was supposed to be inde-

Lighting as a service

A company named Enlighted is offering a Global Energy Optimisation programme that sees it offering lighting solutions as a service to companies. Enlighted will take care of the upgrade costs and capital expenditure involved initially and will later collect a set rate from its customers depending on the saved electricity. The company guarantees savings of 10 to 20 per cent on the energy bill.